

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – SUB-STRAND 1.1 REAL NUMBERS: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: Why can we never write some everyday numbers down completely, and how do mathematicians and Kenyan professionals work accurately with numbers that go on forever? Use what you learned in Lessons 1 to 8 to answer the five sections below. Refer to the Ugali Pot Problem from Gikomba Market: a sufuria with a diameter of 28 cm, where $28 \times \pi = 87.96459430\dots$ on the calculator and the daughter cut her string to 88 cm. Write in full sentences, show every step of any calculation, and use the correct mathematical words. Use your own words and the evidence from our class work.

Part 1: Which numbers can be written down exactly?

Look at the table of everyday Kenyan numbers.

Number	Where it comes from
60.50	Price of 1 kg of unga (Ksh)
160	Distance from Nairobi to Nakuru (km)
34.5	Temperature in Kisumu (°C)
$\frac{1}{3}$	Share of a kilo of unga
$\sqrt{2}$	Diagonal of a square with sides of 1 m
π	Used for the circumference of the sufuria

(a) Classify each number as rational or irrational. (b) Describe what the decimal form of each type of number looks like. (c) State the test that decides whether a number is rational.

Write your answer here:

Part 2: Why the sufuria circumference cannot be written completely

The sufuria has a diameter of exactly 28 cm, which is a nice whole number. Explain why its circumference cannot be written as an exact fraction or a terminating decimal. In your answer, name the type of each number in the calculation $28 \times \pi$ and say what kind of number the product is. Also compare with $28 \times 3 = 84$.

Write your answer here:

Part 3: Working accurately with numbers that never end

The daughter in Gikomba rounded her answer to 88 cm. (a) Explain what she changed about the type of number when she did this. (b) Show what happens if she uses $\frac{22}{7}$ in place of π , and say what type of number you get. (c) Explain why a civil engineer in Kisumu sizing a water-tank rim should be careful about when to round, using what you learned about BODMAS and rounding.

Write your answer here:

Part 4: Measuring how wrong the approximation is

Use $28\pi = 87.96459430\dots$ to find the error in the daughter's 88 cm string. (a) Write the expression for the error using absolute value, and calculate it to 3 decimal places. (b) Is the error a rational or an irrational number? Explain using what you learned about combining rational and irrational numbers. (c) Explain why absolute value is a suitable tool for stating the size of an error.

Write your answer here:

Part 5: Final explanation of the driving question

Write a clear final explanation that answers the driving question: Why can we never write some everyday numbers down completely, and how do mathematicians and Kenyan professionals work accurately with numbers that go on forever? Use at least four ideas from the lessons, including the sufuria in Gikomba Market, the real number system (rational and irrational numbers), the number line, and the use of absolute value. Mention at least one Kenyan professional.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Classifying real numbers	Correctly classifies all numbers as rational or irrational, with the p/q test and decimal behaviour (terminating, repeating, non-terminating non-repeating) explained accurately.	Classifies most numbers correctly and mentions either the p/q test or the decimal behaviour.	Classifies some numbers correctly but gives little or no reason, or confuses recurring decimals with irrational numbers.
Explaining the phenomenon	Clearly explains that 28π is irrational because it is a rational multiplied by the irrational π , and links this to the endless decimal and the gap or overlap in the string.	States that π is irrational and links it to the endless digits, with a partial explanation of the product.	Describes the calculator result but does not link it to irrationality.
Accuracy and approximation	Explains the effect of rounding or using $22/7$, shows $28 \times 22/7 = 88$, and gives the strategy of carrying exact values and	Explains the approximation idea and gets most calculations correct, but the rounding strategy is not fully explained.	Calculations are incomplete or incorrect, and rounding is not explained.

	rounding only at the end.		
Absolute value and error	Correctly calculates $ 88 - 28\pi \approx 0.035$ cm, states it is irrational with a reason, and explains why absolute value suits error measurement.	Calculates the error correctly but gives an incomplete reason for its type or for the use of absolute value.	Does not use absolute value correctly, or the error calculation is incorrect.
Final explanation and use of evidence	Gives a coherent explanation of the driving question using at least four lesson ideas and Kenyan contexts, with accurate mathematical vocabulary.	Gives a mostly clear explanation using two or three ideas from the lessons.	Gives a general or incomplete answer with few ideas from the lessons or little use of Kenyan contexts.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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